
MS130 Datasheet

Overview

Cisco Meraki MS130 switches provide Layer 2 access switching, ideal for branch and campus deployments. The MS130 series features a variety of options designed to meet the diverse needs of branch and campus deployments.

With 10 different models, capable of providing up to 740W of power over a variety of port densities, including SFP+ capable uplinks, the MS130 line is fully ready to support future wireless infrastructure deployments across a variety of different environments.



Features

- Managed via Cisco Meraki Dashboard
- Remote Packet Capture Tools via Meraki Dashboard
- Automatic Firmware upgrades
- SNMP/Syslog Integration
- IPv4/6 ACL support
- 802.1Q VLAN tagging
- DHCP Snooping
- 802.1X Authentication
- 10/100/1000 Mbps RJ45
- 100M/1G/2.5G mGbE RJ45
- 4x 10 Gbe SFP+ models available
- 802.3at PoE up to 30W per port

Configuration

The basic initial configuration of the MS130 is just as simple as any other model of MS switch. The links below provide additional information and instructions relating to each step in getting the device setup and configured for the first time.

1. [Claim the device to an Organization on the Meraki Dashboard](#)
 - a. If a Dashboard Organization does not yet exist, [Create one](#)
2. [Add the device to a Dashboard Network](#)
 - a. If a Network does not yet exist, [Create one first](#)
3. Physically connect the device to the local network
 - a. Connect one of the RJ45 or SFP/SFP+ ports to existing infrastructure to provide a temporary uplink
 - b. Power on the device and let it check in to the Dashboard
 - c. If necessary, configure a Static IP through the [Local Status Page](#) to allow it to communicate with the Meraki Dashboard.
4. Allow the device to complete check-in and perform any initial firmware upgrades
5. Finish configuring the device from the Meraki Dashboard
 - a. [Manage local VLANs / Port configuration](#)

Technical Breakdown

Hardware Breakdown

MS130 Models

	MS130-8	MS130-8P	MS130-8P-I	MS130-8X	MS130-12X
1Gbe RJ45	8	8	8	6	8
mGbe RJ45 (100M/1G/2.5G)	-	-	-	2 x 2.5G	4 x 2.5G
1 Gbe SFP	2	2	2	-	-
10GbE SFP+	-	-	-	2	2
Dedicated Mgmt Interface	-	-	-	-	-
PoE Type	-	802.3at	802.3at	802.3at	802.3at
PoE Port Budget	-	30W	30W	30W	30W
PoE Switch Budget	-	120W	120W	120W	240W
Power Input	12VDC, 2.5A	54VDC, 2.78A	100-240V~, 2.8A-1.2A, 50-60Hz	54VDC, 2.78A	54VDC, 5.56A

	MS130-8	MS130-8P	MS130-8P-I	MS130-8X	MS130-12X	
Power Load (idle/max)	8W/8W	8W/128W	8W/128W	16W/136W	19W/260W	
Operating Temperature	32°F - 113 °F 0°C - 45°C	32°F - 113 °F 0°C - 45°C	32°F - 113 °F 0°C - 45°C	32°F - 113 °F 0°C - 45°C	32°F - 113 °F 0°C - 45°C	
Storage and Transportation Temperature	-4°F - 158 °F -20°C - 70°C	-4°F - 158 °F -20°C - 70°C	-4°F - 158 °F -20°C - 70°C	-4°F - 158 °F -20°C - 70°C	-4°F - 158 °F -20°C - 70°C	
Humidity	5% to 95%	5% to 95%	5% to 95%	5% to 95%	5% to 95%	
Mounting	Desktop	Desktop	Desktop	Desktop	Desktop	
	Integrated Wall mount	Integrated Wall mount	Integrated Wall mount	Integrated Wall mount	Integrated Wall mount	
Switching Capacity	20 Gbps	20 Gbps	20 Gbps	62 Gbps	76 Gbps	
Power Supply	External	External	Internal	External	External	
Fan Operation	Fanless	Fanless	Fanless	Fixed Internal	Fixed Internal	
Dimensions (h x w x d)	1.1 x 8.74 x 6in	1.1 x 8.74 x 6in	1.75 x 9 x 8.58in	1.75 x 9 x 8.58in	1.75 x 9 x 8.58in	
	(2.8 x 22.2 x 15cm)	(2.8 x 22.2 x 15cm)	(4.4 x 23 x 23cm)	(4.4 x 23 x 23cm)	(4.4 x 23 x 23cm)	
Weight	1.94 lb (0.88 kg)	1.94 lb (0.88 kg)	2.86 lb (1.3 kg)	2.12 lb (0.96 kg)	2.34 lb (1.06 kg)	
	MS130-24	MS130-24P	MS130-24X	MS130-48	MS130-48P	MS130-48X
1Gbe RJ45	24	24	18	48	48	40
mGbe RJ45 (100M/1G/2.5G)	-	-	6 x 2.5G	-	-	8 x 2.5G
1Gbe SFP	4	4	-	4	4	-
10Gbe SFP+	-	-	4	-	-	4
Dedicated Mgmt Interface	1	1	1	1	1	1
PoE Type	-	802.3at	802.3at	-	802.3at	802.3at
PoE Port Budget	-	30W	30W	-	30W	30W
PoE Switch Budget	-	370W	370W	-	740W	740W
Power Input	100-240V~, 1.5-0.85A,	100-240V~, 8A-4A, 50-60Hz	100-240V~, 8A-4A, 50-60Hz	100-240V~, 1.5-0.85A,	100-240V~, 12-6A, 50-60Hz	100-240V~, 12-6A, 50-60Hz

	MS130-24	MS130-24P	MS130-24X	MS130-48	MS130-48P	MS130-48X
	50-60Hz			50-60Hz		
Power Load (idle/max)	15W/15W	32W/406W	50W/421W	28W/28W	49W/803W	60W/808W
Operating Temperature	32°F - 113 °F 0°C - 45°C	32°F - 113 °F 0°C - 45°C	32°F - 113 °F 0°C - 45°C	32°F - 113 °F 0°C - 45°C	32°F - 113 °F 0°C - 45°C	32°F - 113 °F 0°C - 45°C
Storage and Transportation Temperature	-4°F - 158 °F -20°C - 70°C	-4°F - 158 °F -20°C - 70°C	-4°F - 158 °F -20°C - 70°C	-4°F - 158 °F -20°C - 70°C	-4°F - 158 °F -20°C - 70°C	-4°F - 158 °F -20°C - 70°C
Humidity	5% to 95%	5% to 95%	5% to 95%	5% to 95%	5% to 95%	5% to 95%
Mounting	Integrated 1U Rack Mount	Integrated 1U Rack Mount	Integrated 1U Rack Mount	Integrated 1U Rack Mount	Integrated 1U Rack Mount	Integrated 1U Rack Mount
Switching Capacity	56 Gbps	56 Gbps	146 Gbps	104 Gbps	104 Gbps	200 Gbps
Power Supply	Fixed Internal	Fixed Internal	Fixed Internal	Fixed Internal	Fixed Internal	Fixed Internal
Fan Operation	Fanless	Fixed Internal	Fixed Internal	Fanless	Fixed Internal	Fixed Internal
Dimensions	1.73 x 17.32 x 10in	1.73 x 17.32 x 10in	1.73 x 17.32 x 10in	1.73 x 17.32 x 10in	1.73 x 17.32 x 13.4in	1.73 x 17.32 x 13.4in
	(4.4 x 44 x 25cm)	(4.4 x 44 x 25cm)	(4.4 x 44 x 25cm)	(4.4 x 44 x 25cm)	(4.4 x 44 x 34cm)	(4.4 x 44 x 34cm)
Weight	7.19 lb (3.26 kg)	8.91 lb (4.04 kg)	9.37 lb (4.25 kg)	8.14 lb (3.69 kg)	12.13 lb (5.5 kg)	12.59 lb (5.71 kg)

Whats In the Box

Model	Included
MS130-8, 8P, 8X, 12X	MS130 switch, AC Power Supply
MS130-8P-I, MS130-24, 24P, 24X, 48, 48P, 48X	MS130 switch

Region-specific power cords are not included in the box*. Order the appropriate power cord separately:

- MA-PWR-CORD-US
- MA-PWR-CORD-EU
- MA-PWR-CORD-UK
- MA-PWR-CORD-CN
- MA-PWR-CORD-IN
- MA-PWR-CORD-BR
- MA-PWR-CORD-TW

- MA-PWR-CORD-AU
- MA-PWR-CORD-AR
- MA-PWR-CORD-JP



**1 MA-PWR-CORD-US is included automatically with US orders only*



Rack mount screws are included with the MS130-24, 24P, 24X, 48, 48P, 48X models.

Accessories

Power Supplies

Description	Accessory	Supported models
150 W power adapter	MA-PWR-150WAC-ADP	MS130-8X and MS130-8P
300 W power adapter	MA-PWR-300WAC-ADP	MS130-12X
30 W power adapter	MA-PWR-30WAC	MS130-8

SFP Modules

The following SFP/Fiber transceivers are supported

SFP Models

- MS130-8, 8P, 8P-I
- MS130-24, 24P
- MS130-48, 48P

Supported Modules

- MA-SFP-1GB-SX
- MA-SFP-1GB-LX10
- MA-SFP-1GB-TX

Supported Modules

SFP+ Models

- MS130-8X
- MS130-12X
- MS130-24X
- MS130-48X

- MA-SFP-1GB-SX
- MA-SFP-1GB-LX10
- MA-SFP-1GB-TX
- MA-SFP-10GB-SR
- MA-SFP-10GB-LR
- MA-SFP-10GB-ER
- MA-CBL-TA-1M
- MA-CBL-TA-3M



* For more information regarding the SFP modules, see the following Cisco Meraki datasheets:



Troubleshooting

The MS uses LEDs to inform the user of the device's status. Functions are described below, from left to right. For fixed Ethernet ports, the status LED is on the top left or bottom right depending on port orientation. There is also a traffic LED which flashes orange as traffic is sent/received through that port.

Function	LED Status	Meaning
Power	Solid orange	Switch is unable to connect to the Meraki cloud
	Flashing white	Firmware upgrade in process
	Solid white	Switch is fully operational and connected to the Meraki cloud
	Off	Switch does not have power
Switch Ports	Off	Port is operating at less than full speed. For example: • 10/100M on a 1GE port • 100M/1GE on 2.5GE ports • 1G on SFP+ ports
	Solid green	Port is operating at full speed • 1GE on 1GE ports • 2.5GE on 2.5GE ports • 10GE on SFP+

Common Troubleshooting

My device is connected to the network but not checking in to the Meraki cloud or shows a solid Orange LED.

Confirm that the device is powered on and has a valid IP address that is able to access the Internet. Use the Local Status Page to get more information about the connectivity status of the device such as if it can successfully reach the Local Gateway, Internet, and/or Meraki Cloud servers. If necessary, contact Meraki Support for additional assistance.

My Status LED is blinking WHITE

A blinking WHITE Status LED indicates that the device is in contact with the Dashboard Cloud servers and is performing a firmware update. This can sometimes take 20-45 minutes or more to complete depending on hardware and other factors.

My Status LED is blinking ORANGE

The device is not able to successfully communicate with the Dashboard Cloud servers or there may be a hardware issue with the device. Check the Local Status Page of the device to confirm the status and reach out to Meraki Support for further troubleshooting.

Event Log

The most common Event Log messages and their meaning are listed below.

- **Port STP change:** Indicates the STP state of the port has changed, lists the relevant port number, previous, and new states. Typically accompanied by a 'Port status change' event.
- **Port status change:** Indicates the link state of the port has changed, lists the relevant port number, old, and new state. Always accompanied by a 'Port STP change' event.
- **SFP module inserted/removed:** Indicates that an SFP module was either inserted or removed, includes SFP module information for inserted events and always lists the relevant port number.

Licensing

MS130 license structure includes two feature tiers: Enterprise and Advanced. The Enterprise license is available in 1, 3, 5, 7, and 10 year terms. The Advanced License is available in the same terms, except for 7 and 10 years. All MS130s in an organization must have Enterprise or Advanced licenses - they cannot be mixed.

Model	License	Description
MS130-8		
MS130-8P	LIC-MS130-CMPT-xY	MS130 compact switch (8/12) Enterprise License and support
MS130-8P-I	LIC-MS130-CMPTA-xY	MS130 compact switch (8/12) Advance License and Support
MS130-8X		
MS130-12X		
MS130-24		
MS130-24P	LIC-MS130-24-xY	MS130 24 ports Enterprise License and support
MS130-24X	LIC-MS130-24A-xY	MS130 24 ports Advance License and Support
MS130-48		
MS130-48P	LIC-MS130-48-xY	MS130 48 ports Enterprise License and support
MS130-48X	LIC-MS130-48A-xY	MS130 48 ports Advance License and Support



In the Co-term licensing model (most existing Organizations), an Organization must either have all MS130 Enterprise or all MS130 Advanced licenses - they cannot be mixed. **Additionally, a co-term organization cannot mix enterprise and advanced licenses on any switches that have both enterprise and advanced licenses.**

For example, if an organization has existing MS390/C9300/MS150s with enterprise licenses, they cannot add MS130's with advanced licenses in that organization - they must be added with enterprise licenses.

If the organization has existing MS390/C9300/MS150s with advanced licenses, they cannot add MS130's with enterprise licenses in that organization - they must be added with advanced licenses.

In the Per-device licensing model, a mix of Enterprise and Advanced can be added to a single Organization, but certain features may require all devices in a Organization to have Advanced licenses, e.g. Adaptive Policy.

MS130 Advanced License only provides [Adaptive policy](#) as an additional feature.

For more information about the differences between the Advanced and Enterprise licenses see the below links:

- [Meraki Co-Termination Licensing Overview](#)
- [MS Family Datasheet](#)

The MS130 is also available as [Subscription Licensing](#):

Model	License
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MS130-8

MS130-8P MS100 Small

MS130-8P-I Essentials: LIC-MS-100-S-E

MS130-8X Advantage: LIC-MS-100-S-A

MS130-12X

MS130-24 MS100 Medium

MS130-24P Essentials: LIC-MS-100-M-E

MS130-24X Advantage: LIC-MS-100-M-A

MS130-48 MS100 Large

MS130-48P Essentials: LIC-MS-100-L-E

MS130-48X Advantage: LIC-MS-100-L-A MS120

For more information on licensing, refer to [Meraki Licensing Models article](#).

MTBF Rating

Model	MTBF at 25°C (in hours)
MS130-8	1,927,507
MS130-8P	563,569
MS130-8P-I	608,122
MS130-8X	1,042,108
MS130-12X	682,400
MS130-24	789,842
MS130-24P	768,152
MS130-24X	677,546
MS130-48	425,296
MS130-48P	309,196
MS130-48X	396,536